

Software Test Plan (STP) - Vehicle Parking System (VPS)

Project: Vehicle Parking System (VPS)

Version: 1.0

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Date: 16-09-2026

Status: Draft / Approved

1. Introduction

Purpose

This document defines the software test plan for the Vehicle Parking System (VPS) v1.0. It outlines the testing objectives, scope, strategy, resource allocations, schedule, and responsibilities for verifying system functionality, performance, security, and hardware integrations.

Scope

Testing covers:

- Vehicle entry/exit automation at gate lanes via Automatic Number Plate Recognition (ANPR) and QR tickets.
- Real-time slot vacancy detection via IoT sensors and multi-category slot allocation.
- Dynamic fare computation, grace period enforcement, surge pricing, and cashless payment integration (UPI, Cards, Wallets).
- 48-hour slot pre-booking and auto-cancellation workflows.

- Administrative rate management, RBAC, audit logging, exportable analytics reporting, and 72-hour edge gate offline queuing resilience.

Excluded: Internal payment switch banking implementations and physical towing enforcement.

References

- VPS Software Requirements Specification (SRS) v1.0 (Date: 09-09-2026)
- VPS Software Architecture and Design Specification (SAD) v1.0 (Date: 09-09-2026)
- PCI-DSS Compliance Guidelines & TLS 1.3 Communication Standards

Definitions

- **VPS:** Vehicle Parking System
- **ANPR:** Automatic Number Plate Recognition
- **VMS:** Variable Message Sign (Digital LED display boards)
- **RTM:** Requirements Traceability Matrix
- **POS:** Point of Sale Terminal

2. Test Items

- **Ingress Automation Module:** ANPR camera OCR capture, ticket generation, ground loop safety checks, and barrier relay control.
- **Slot Allocation Engine & Sensor Hub:** Slot assignment algorithms, ultrasonic occupancy sensors, and entrance VMS display board updater.
- **Tariff & Billing Engine:** Dynamic tiered hourly fare calculation, 15-minute grace period enforcement, coupon discounts, and surge pricing.
- **Payment Gateway & Egress Module:** Cashless UPI/Card processing, receipt generation, and exit barrier release.
- **Reservation Module:** 48-hour advance online booking and 30-minute no-show auto-cancellation scheduler.

- **Admin Console & Analytics Engine:** Dynamic tariff layout configuration, multi-factor authentication (MFA), and exportable PDF/CSV revenue/audit reports.
- **Edge Gate Controller:** Local SQLite store-and-forward queue for offline operations.

3. Features to be Tested

Features mapped to SRS requirement IDs:

- **VPS-F-001:** ANPR license plate capture within 1.5 seconds ($\geq 95\%$ accuracy).
- **VPS-F-002:** Encrypted QR ticket issuance within ≤ 2 seconds.
- **VPS-F-003 & VPS-F-004:** Ingress boom barrier response ($\leq 500\text{ms}$) and ground loop anti-tailgating safety halt.
- **VPS-F-005 to VPS-F-007:** Vacant slot allocation by vehicle class, real-time sensor LED toggling ($\leq 2\text{s}$), and entrance VMS count update ($\leq 1\text{s}$).
- **VPS-F-008 to VPS-F-010:** Dynamic fare calculation, 15-minute zero-charge grace period, and surge/promo coupon logic.
- **VPS-F-011 to VPS-F-014:** Exit QR/plate scan ($\leq 1\text{s}$), cashless payment settlement ($\leq 3\text{s}$), barrier release, and GST tax invoice generation.
- **VPS-F-015 & VPS-F-016:** 48-hour advance slot reservation and 30-minute no-show slot release cron job.
- **VPS-F-017 & VPS-F-018:** Admin dashboard configuration and CSV/PDF report export ($\leq 5\text{s}$).
- **VPS-NF-001:** Ingress transaction response time ≤ 3 seconds (90th percentile under normal load).
- **VPS-NF-002 & VPS-NF-003:** System availability (99.9%) and scaling (10,000 slots, 50 gate lanes, $< 200\text{ms}$ latency at 500 req/sec).
- **VPS-NF-005:** 72-hour autonomous edge gate offline queuing and auto-syncing upon reconnection.

4. Features Not to be Tested

- Core banking switch clearing routines (assumed verified by payment gateway partners Razorpay/Stripe).
- Physical structural durability and weatherproofing of outdoor enclosures.
- Physical towing enforcement and wheel clamping mechanisms.

5. Test Approach / Strategy

5.1 Levels

- **Unit Testing:** Independent validation of core calculation algorithms (TariffEngine fare rules, HMAC-SHA256 signature generation).
- **Integration Testing:** Verification of hardware-to-software interfaces (ANPR camera stream, Relay barrier controllers, Ultrasonic sensor telemetry via MQTT, Payment Webhook endpoints).
- **System Testing:** End-to-end testing of complete ingress, slot occupancy, billing, payment, and egress workflows.
- **Acceptance Testing (UAT):** Validation by facility managers and parking attendants against real-world scenario checklists.

5.2 Types

- **Functional Testing:** Validating core features against acceptance criteria (AC-VPS-F-001 through AC-VPS-F-018).
- **Regression Testing:** Automated regression suite executed on each build to ensure core gate workflows remain unaffected.
- **Performance & Load Testing:** Simulating high traffic (500 req/sec across 50 simulated gate lanes) using JMeter.
- **Resilience & Fault Tolerance Testing:** Simulating WAN network disconnects to test 72-hour SQLite edge gate queuing and auto-sync behavior.

5.3 Entry & Exit Criteria

- **Entry Criteria:** SRS and Design Specifications finalized and approved; stable software build deployed; hardware stubs and sandbox APIs configured.
- **Exit Criteria:** 100% of planned test cases executed; zero critical/high defects open; 100% pass rate for security and performance NFRs.

5.4 Security Validation

- **VPS-SR-001:** Mandatory TLS 1.3+ verification across all communication channels.
- **VPS-SR-002:** Verification of password hashing (Argon2id/bcrypt) and AES-256 encryption at rest for PII data.
- **VPS-SR-003:** Authentication enforcement testing for RBAC and MFA on admin portals.
- **VPS-SR-004:** HMAC-SHA256 signature verification on entry QR codes to detect counterfeit tickets.
- **VPS-SR-005:** Audit log integrity testing for immutable logging of manual overrides and tariff edits.

6. Test Environment

Category	Items
Hardware	Edge Gate Controllers (Linux IoT), ANPR cameras, Boom barriers, Ground loops, Ultrasonic sensors, QR thermal printers, VMS LED boards, POS terminals.
Software	VPS Application Server v1.0, Edge Gate OS, PostgreSQL DB, Redis, SQLite Store-and-Forward Queue.

Category	Items
Tools	Selenium/Cypress (UI), Postman/REST-Assured (API), Apache JMeter (Load), Jira (Defect Management), MQTTX (IoT).
Test Data	Dummy plate numbers, pre-configured test accounts/cards, mock UPI webhooks, simulated ultrasonic sensor streams.

7. Test Schedule

Milestone	Target Date
Test Case Design & RTM Finalization	20-Sep-2026
Test Environment & Hardware Stub Setup	23-Sep-2026
Functional & Integration Test Execution	25-Sep-2026
Performance, Security & Resilience Testing	02-Oct-2026
User Acceptance Testing (UAT)	05-Oct-2026
Final Test Sign-off	10-Oct-2026

8. Test Deliverables

- Software Test Plan (this document)
- Test Cases Suite (Manual & Automated)
- Test Automation Scripts & JMeter Load Test Profiles
- Test Data Sets & Edge Simulation Scripts

- Defect Reports & Requirements Traceability Matrix (RTM)
- Final Test Summary Report

9. Roles and Responsibilities

Role	Name	Primary Responsibility
QA Lead ▾	Angelo Arakal	Prepare test plan, coordinate execution, manage RTM and triage.
Test Engineer ▾	Himansh Ahuja	Design and execute automated scripts for Ingress, Billing, and API.
Test Engineer ▾	Shweta Mailaragouda Patil	Design and execute hardware integration, IoT, and performance tests.
Software Engineer ▾	Akarsh Parashuram	Environment setup, bug triage, and applying fixes.
Evaluator ▾	Prof. Pavithra	Review test outcomes and approve sign-off readiness.

10. Risks and Mitigation

- **Risk:** Physical gate hardware unavailability during initial phases.
 - *Mitigation:* Develop mock hardware stubs and MQTT payload simulators for barriers and sensors.
- **Risk:** Flaky ANPR OCR recognition due to lighting or camera angle variations.
 - *Mitigation:* Use standard image datasets representing diverse weather/lighting conditions.

- **Risk:** Third-party payment gateway sandbox latency or downtime.
 - *Mitigation:* Implement mock payment webhooks to decouple billing logic validation.

11. Assumptions & Dependencies

- PostgreSQL database and Redis memory cache instances will be pre-configured in the staging environment.
- Payment Gateway sandbox APIs (Razorpay/Stripe) will be operational during exit flow execution.
- ANPR camera feeds and edge gate controllers comply with standard RTSP and Wiegand/RS-485 protocols.

12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment or edge database is unavailable for > 4 consecutive hours.
- Critical blocking defects disrupt more than 30% of primary ingress/egress test cases.

Resume testing if:

- Blocking defects are resolved and verified via smoke test build.
- Test environment and network connectivity are restored and stabilized.

13. Test Case Management & Traceability

Requirement ID	Requirement Description	Test Case ID
VPS-F-001	ANPR Plate Capture	TC-ENT-01

Requirement ID	Requirement Description	Test Case ID
VPS-F-002	QR Ticket Generation	TC-ENT-02
VPS-F-003	Ingress Barrier Signal	TC-ENT-03
VPS-F-004	Ground Loop Safety Check	TC-ENT-04
VPS-F-005	Nearest Vacant Slot	TC-SLOT-01
VPS-F-008	Dynamic Fare Computation	TC-BILL-01
VPS-F-012	Cashless Payments	TC-PAY-02
VPS-F-015	48h Advance Slot Pre-Booking	TC-RES-01
VPS-NF-001	Ingress Latency	TC-Perf-01
VPS-NF-005	Edge Offline Resilience	TC-Fault-01
VPS-SR-001/5	Security Requirements	TC-SEC-01 to 05

14. Test Metrics & Reporting

Metrics Collected

- Percentage of test cases executed, passed, failed, or blocked.
- Total defect count by severity (Critical, High, Medium, Low).
- Defect resolution aging and reopen rate.
- Requirement coverage percentage (100% target for all 18 FRs and 5 NFRs).

Reports

- Daily execution status report.

- Weekly defect triage summary.
- Final Test Summary and Sign-off Report.

15. Approvals

QA Lead (Angelo Arakal): _____

Course Coordinator (Prof. Sheela Devi): _____

Project Evaluator / Guide (Prof. Pavithra): _____